COMPUTER SCIENCE BOOK

Test Preparation - Notes & MCQs

PST / JEST - SST - SPSC - FPSC Screening Tests

Computer Basics - Hardware - MS Office - Internet - Number Systems - Programming

Short Notes - Key Facts & Charts - 183 Practice MCQs with Answer Keys

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How to use this book: Har chapter mein pehle short notes parhein, phir key facts wala khaaka dekhein, aur aakhir mein practice MCQs khud hal karein. Jawab chapter ke aakhir mein diye gaye Answer Key se check karein. Computer ke sawal zyada tar joron (device-kaam, shortcut-key, full form) aur farq (RAM-ROM, compiler-interpreter, browser-search engine) par bante hain - unhein jor aur farq ke saath yaad karein, poora chapter asan ho jayega.

Chapter 1

Computer Basics aur Generations

Har computer test ki shuruaat yahin se hoti hai - computer kya hai, data aur information mein kya farq hai, aur computer ki paanch generations kon si hain. PST, JEST, SPSC aur FPSC ke har paper mein is chapter se sawal aate hain.

Notes - Asaan Alfaaz Mein

- **Computer kya hai:** computer ek **electronic machine** hai jo ham se data leti hai (input), us par diye gaye usoolon ke mutabiq kaam karti hai (processing), use mehfooz rakhti hai (storage) aur nateeja deti hai (output). Is poore cycle ko **IPO cycle** kehte hain - Input, Process, Output. Computer khud soch nahi sakta; jo program (instructions) diya jaye, wahi karta hai - is liye kehte hain ke computer ka apna IQ zero hota hai.
- **Data aur Information ka farq: Data** kache (raw) facts aur figures hain - masalan numbers, naam, marks. Jab data ko process karke maani-daar (meaningful) shakal di jaye to woh **Information** ban jata hai. Masalan "85" sirf data hai, lekin "Ali ne Maths mein 85 marks liye" information hai. Test mein ye farq bar bar poocha jata hai.
- **Computer ki khususiyat (characteristics):** (1) **Speed** - lakhon kaam ek second mein; (2) **Accuracy** - agar input sahi ho to nateeja ghalti ke baghair; (3) **Storage** - bohat bara data mehfooz rakhna; (4) **Diligence** - thake baghair ek jaisa kaam bar bar karna; (5) **Versatility** - ek hi machine se bohat se mukhtalif kaam. Insaan thak jata hai, computer nahi thakta - ye us ki bari taqat hai.
- **First Generation (1940s-1950s):** in computers mein **vacuum tubes** istemal hue. Ye bohat bare, garam hone wale aur bijli zyada khane wale thay. Mashhoor misalein: **ENIAC** (1946) - pehla general-purpose electronic computer, aur **UNIVAC** - pehla commercial computer. Language sirf machine language thi.
- **Second Generation (1950s-1960s):** vacuum tubes ki jagah **transistors** aa gaye. Computer chhota, sasta, tez aur kam garam hone laga. Is dor mein assembly language aur shuru ki high-level languages (FORTRAN, COBOL) aayen.
- **Third Generation (1960s-1970s):** is dor ki pehchan **Integrated Circuit (IC)** hai - ek chhoti si chip par bohat se transistors. Computer aur chhota zehn tez hua, keyboards aur monitors aam hue, aur operating systems behtar hue.
- **Fourth Generation (1970s se aaj tak):** is ki pehchan **Microprocessor** hai - puri CPU ek hi chip par. Isi dor mein Personal Computer (PC), laptop, GUI aur internet aam hue. Technology ko VLSI (Very Large Scale Integration) kehte hain.
- **Fifth Generation (present aur future):** ye generation **Artificial Intelligence (AI)** aur **ULSI** technology par mabni hai - computer insaan ki tarah seekhne, awaz pehchannay aur faislay mein madad karne ki koshish karta hai. Voice assistants aur expert systems is ki misalein hain.
- **Tareekh ke aham naam: Charles Babbage** ko "Father of Computer" kehte hain - unhone Analytical Engine ka design banaya. **Ada Lovelace** ko duniya ka pehla computer programmer mana jata hai. Ginti ka sab se purana auzaar **Abacus** hai jo China se jura hai. Computers ki aqsaam: **Analog** (lagatar miqdaar - jese thermometer reading), **Digital** (numbers par kaam), **Hybrid** (dono ki khusosiyat). Size ke lehaz se: **Supercomputer** sab se tez (mosam aur research ke liye), phir Mainframe, Minicomputer aur Microcomputer (hamara PC).

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Paanch Generations - ek nazar mein

1st: Vacuum Tubes - ENIAC, UNIVAC - machine language.

2nd: Transistors - chhota aur tez - FORTRAN, COBOL.

3rd: Integrated Circuits (IC) - keyboard, monitor, OS.

4th: Microprocessor - PC, laptop, GUI, internet.

5th: Artificial Intelligence (ULSI) - voice, expert systems.

Box 2: Computer ki aqsaam - size ke lehaz se

Supercomputer: sab se tez; mosam ki peshgoi, research.

Mainframe: bare idaray, bank - ek saath hazaron users.

Minicomputer: darmiyani; chhoti companies ke liye.

Microcomputer: ek user - PC, laptop, tablet, mobile.

Aam Ghaltiyan - Common Mistakes

- ENIAC pehla general-purpose electronic computer hai, UNIVAC pehla COMMERCIAL computer - dono ko ulta karna aam ghalti hai.
- Data aur information ek cheez nahi - data kacha hota hai, information process ke baad maani-daar banta hai.
- IC third generation ki pehchan hai, microprocessor fourth ki - ye jor test mein ulta karke poocha jata hai.
- Computer "soch" nahi sakta - us ka apna IQ zero hota hai; woh sirf diye gaye program par chalta hai.

Practice MCQs

1. Who is known as the Father of the Computer?

(a) Charles Babbage (b) Alan Turing (c) Bill Gates (d) John von Neumann

2. The first general-purpose electronic computer was:

(a) UNIVAC (b) ENIAC (c) EDSAC (d) Abacus

3. First generation computers used:

(a) Transistors (b) Integrated circuits (c) Vacuum tubes (d) Microprocessors

4. Second generation computers used:

(a) Vacuum tubes (b) Microprocessors (c) Optical fibres (d) Transistors

5. Third generation computers are known for the use of:

(a) Vacuum tubes (b) Transistors only (c) Integrated Circuits (IC) (d) Artificial intelligence

6. The main technology of fourth generation computers is:

(a) Microprocessor (b) Vacuum tube (c) Transistor (d) Punch cards

7. Fifth generation computers are based on:

(a) Vacuum tubes (b) Transistors (c) Punch card machines (d) Artificial Intelligence and ULSI

8. Raw facts and figures are called:

(a) Information (b) Data (c) Software (d) Output

9. Data becomes information when it is:

(a) Stored on a disk (b) Processed and made meaningful (c) Printed on paper (d) Deleted from memory

10. The working cycle of a computer is called:

(a) Data cycle (b) Program cycle (c) Machine round (d) IPO cycle (Input-Process-Output)

11. The earliest known tool used for counting and calculation is:

(a) Calculator (b) ENIAC (c) Abacus (d) Slide rule machine

12. Who is considered the first computer programmer in the world?

(a) Ada Lovelace (b) Charles Babbage (c) Alan Turing (d) Ada Byron's teacher

13. The fastest and most powerful type of computer is:

(a) Mainframe computer (b) Minicomputer (c) Microcomputer (d) Supercomputer

14. Computers that work on continuous physical quantities such as voltage or temperature are called:

(a) Digital computers (b) Analog computers (c) Hybrid computers (d) Micro computers

15. A computer which combines the features of both analog and digital computers is called:

(a) Hybrid computer (b) Personal computer (c) Mainframe computer (d) Supercomputer

16. ENIAC was announced/built in the year:

(a) 1936 (b) 1951 (c) 1946 (d) 1961

17. Which of the following is NOT a characteristic of a computer?

(a) It can think and take decisions on its own (b) Speed (c) Accuracy (d) Diligence

Answer Key - Chapter 1

1. (a) 2. (b) 3. (c) 4. (d) 5. (c) 6. (a) 7. (d) 8. (b) 9. (b) 10. (d) 11. (c) 12. (a) 13. (d) 14. (b) 15. (a) 16. (c) 17. (a)

Try Yourself - Quick Drills

1. Father of Computer kis ko kehte hain?
2. First generation mein kon sa hissa istemal hua?
3. Data aur information mein bunyadi farq kya hai?
4. ENIAC ka full form kya hai?
5. Fifth generation kis technology par hai?

Drill Answers: 1. Charles Babbage ko 2. Vacuum tubes 3. Data kacha hota hai, information processed aur maani-daar 4. Electronic Numerical Integrator and Computer 5. Artificial Intelligence (ULSI)

Chapter 2

CPU aur Memory (RAM, ROM, Units)

Computer ka dimagh CPU hai aur us ki yaadasht memory. Units ki tabdeeli (bit, byte, KB, MB, GB) aur RAM-ROM ka farq test mein sab se zyada aane wale sawalon mein se hain.

Notes - Asaan Alfaaz Mein

- **CPU (Central Processing Unit):** CPU ko computer ka **dimagh (brain)** kehte hain. Ye ek chip (microprocessor) par hota hai aur data par asal processing yahi karta hai. Is ke do bare hisse hain: **ALU** aur **CU**. CPU ki raftaar (clock speed) **Hertz** mein napi jati hai - aaj kal GHz (gigahertz) mein hoti hai.
- **ALU (Arithmetic Logic Unit):** ye hisa tamam **hisaab (arithmetic)** - jama, tafreeq, zarb, taqseem - aur **mantiqi (logical)** faislay - bara hai, chhota hai, barabar hai - karta hai. Jab bhi computer koi calculation kare ya do cheezon ka muqabla kare, kaam ALU ka hota hai.
- **CU (Control Unit):** ye hisa poore computer ka **manager** hai - ye tamam hisson ko batata hai ke kab kya karna hai, data kahan se lena hai aur kahan bhejna hai. CU khud hisaab nahi karta; woh hidayat deta aur control karta hai. Is ke ilawa CPU mein chhoti, bohat tez yaadasht **Registers** bhi hote hain.
- **Memory ki aqsaam:** memory do bari aqsaam ki hai - **Primary memory** (CPU ke saath seedha juri hui: RAM, ROM, cache, registers) aur **Secondary memory** (data hamesha ke liye rakhne wali: hard disk, SSD, DVD, USB). Primary memory tez magar mehngi aur chhoti hoti hai; secondary sastti aur bari, magar sust.
- **RAM (Random Access Memory):** ye computer ki **aarzi (temporary)** yaadasht hai. Jo program chal raha ho, woh RAM mein hota hai. Bijli band hote hi RAM ka data **khatam** ho jata hai - is liye RAM ko **volatile memory** kehte hain. RAM zyada ho to computer ek saath zyada program aram se chalata hai. RAM ki do aqsaam: SRAM (tez, cache mein) aur DRAM (aam RAM).
- **ROM (Read Only Memory):** ye **mustaqil (permanent)** memory hai - bijli band hone par bhi data mehfooz rehta hai, is liye ise **non-volatile** kehte hain. Computer ko start karne wala program (**BIOS**) ROM mein hota hai. Iski aqsaam: PROM (ek dafa likhna), EPROM (mita kar dobara likhna, UV light se), EEPROM (bijli ke signal se mitana).
- **Cache Memory:** ye ek bohat chhoti magar **intehai tez** memory hai jo CPU aur RAM ke darmiyan hoti hai. Jo data bar bar istemal ho, woh cache mein rakha jata hai taake CPU ko RAM tak jane ki der na lage. Registers sab se tez, phir cache, phir RAM - ye tartib yaad rakhein.
- **Data ki units (units of storage):** sab se chhoti unit **Bit** hai (0 ya 1). **4 bits = 1 Nibble**, **8 bits = 1 Byte** (ek harf/number rakhne ke liye). Phir seedhi yun hai: **1024 Bytes = 1 KB**, 1024 KB = 1 MB, 1024 MB = 1 GB, 1024 GB = 1 TB, 1024 TB = 1 PB. Yaad rakhein ke computer ki duniya mein har agli unit 1000 nahi, **1024** hoti hai - test mein yahi phasaya jata hai.
- **Booting:** jab computer on karte hain to ROM mein rakha BIOS pehle hardware ko check karta hai, phir operating system ko RAM mein load karta hai. Is poore amal ko **booting** kehte hain. Jab RAM bhar jaye to OS hard disk ka kuch hissa aarzi memory ke tor par istemal karta hai - use **Virtual Memory** kehte hain.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Storage units ki seedhi - chhoti se bari

Bit (0 ya 1) - sab se chhoti unit.

Nibble = 4 bits. Byte = 8 bits = 1 character.

KB = 1024 Bytes. MB = 1024 KB. GB = 1024 MB.

TB = 1024 GB. PB = 1024 TB.

Box 2: RAM aur ROM - saath saath farq

RAM: aarzi (volatile), bijli band = data khatam, chal raha program yahan rehta hai.

ROM: mustaqil (non-volatile), data mehfooz, BIOS yahan hota hai.

RAM par likhna aur parhna dono mumkin; aam ROM sirf parhi jati hai.

Dono primary memory hain - CPU se seedha rabta.

Aam Ghaltiyan - Common Mistakes

- 1 KB = 1024 Bytes hota hai, 1000 nahi - computer units mein 1024 ka usool hai.
- RAM volatile hai (data ujaye jata hai), ROM non-volatile - ye jor ulta karna sab se aam ghalti hai.
- ALU hisaab karta hai, CU control karta hai - dono ke kaam aapas mein badalna ghalat hai.
- Sab se tez memory registers hain, phir cache - hard disk ya RAM ko "sab se tez" likhna ghalat hai.

Practice MCQs

1. Which part of the computer is called its "brain"?

(a) Monitor (b) Hard disk (c) CPU (d) Printer

2. ALU performs which type of operations?

(a) Only printing work (b) Arithmetic and logical operations (c) Storing files permanently (d) Controlling the printer

3. The unit which controls and coordinates all operations of a computer is:

(a) ALU (b) Hard disk (c) Monitor (d) Control Unit (CU)

4. The temporary memory which loses all its data when power is switched off is:

(a) ROM (b) Hard disk (c) DVD (d) RAM

5. RAM stands for:

(a) Random Access Memory (b) Read And Modify (c) Rapid Access Machine (d) Read Only Memory

6. ROM stands for:

(a) Random Only Memory (b) Read Open Machine (c) Read Only Memory (d) Rapid Output Memory

7. How many bits are there in one byte?

(a) 4 bits (b) 8 bits (c) 16 bits (d) 1024 bits

8. One Kilobyte (1 KB) is equal to:

(a) 1000 bytes (b) 1024 bytes (c) 100 bytes (d) 8 bytes

9. A group of 4 bits is called a:

(a) Byte (b) Word (c) Nibble (d) Kilobyte

10. The fastest memory in a computer system is:

(a) Cache memory (b) RAM (c) Hard disk (d) Registers

11. The startup program BIOS is stored in:

(a) ROM (b) RAM (c) Hard disk (d) Cache memory

12. The speed of a CPU is measured in:

(a) Bytes per second (b) Dots per inch (c) Hertz (GHz) (d) Pages per minute

13. Which of the following is a non-volatile memory?

(a) RAM (b) Cache memory (c) Registers (d) ROM

14. Cache memory is located between:

(a) CPU and RAM (b) Monitor and CPU (c) Hard disk and printer (d) Keyboard and monitor

15. Which type of ROM can be erased by ultraviolet light and reprogrammed?

(a) PROM (b) EPROM (c) RAM (d) Cache

16. Which of the following is a primary memory?

(a) Hard disk (b) DVD (c) USB flash drive (d) RAM

17. The smallest unit of data in a computer is:

(a) Byte (b) Nibble (c) Bit (d) Kilobyte

Answer Key - Chapter 2

1. (c) 2. (b) 3. (d) 4. (d) 5. (a) 6. (c) 7. (b) 8. (b) 9. (c) 10. (d) 11. (a) 12. (c) 13. (d) 14. (a) 15. (b) 16. (d) 17. (c)

Try Yourself - Quick Drills

1. CPU ka full form kya hai?
2. RAM ko volatile memory kyun kehte hain?
3. 1 GB mein kitne MB hote hain?
4. Hisaab aur mantiqi kaam kon sa hisa karta hai?
5. BIOS kis memory mein hota hai?

Drill Answers: 1. Central Processing Unit 2. Bijli band hone par us ka data khatam ho jata hai 3. 1024 MB 4. ALU (Arithmetic Logic Unit) 5. ROM mein

Chapter 3

Input, Output aur Storage Devices

Kon si device input hai, kon si output, aur kis ki kitni capacity hai - scanner, OMR, MICR, plotter, DPI aur PPM wale sawal har teacher aur screening test mein zaroor aate hain.

Notes - Asaan Alfaaz Mein

- **Input Devices:** jin cheezon se hum computer ko data aur hidayat dete hain, woh input devices hain - **Keyboard, Mouse, Scanner, Microphone, Webcam, Joystick**. Kuch khaas input devices test ke liye yaad rakhein: **OMR** (Optical Mark Reader) MCQ wali answer sheets check karta hai; **OCR** (Optical Character Reader) likhe hue haroof ko parh kar text banata hai; **MICR** bank ke cheques parhnay ke liye hota hai; **Barcode Reader** dukanon par lines wala code parhta hai.
- **Output Devices:** jo nateeja hamein dikhayein ya dein, woh output devices hain - **Monitor, Printer, Speaker, Plotter**. Monitor ki screen par nazar aane wali sab se chhoti cheez (dot) ko **Pixel** kehte hain; pixel zyada hon to tasveer saaf (resolution behtar) hoti hai. Monitor ki aqsaam: CRT (purana, bhari), LCD aur LED (patle, kam bijli).
- **Printer ki aqsaam:** printers do bare groups mein hain: **Impact printers** kagaz ko chho (strike) kar print karte hain - misal **Dot Matrix** printer (shor wala, carbon copy ke liye acha). **Non-impact printers** chhoye baghair print karte hain - **Inkjet** (siyahi ke chhote qatre) aur **Laser** (sab se tez aur saaf, offices ke liye). Printer ki raftaar **PPM (Pages Per Minute)** mein aur print ki safai/resolution **DPI (Dots Per Inch)** mein napi jati hai.
- **Plotter:** ye khaas output device bare naqshay, building ke blueprints aur engineering drawings print karne ke liye hoti hai. Aam printer se ye bara aur mehanga hota hai - test mein "large maps ke liye kon si device" ka jawab plotter hota hai.
- **Storage Devices - Magnetic: Hard Disk Drive (HDD)** computer ki asal mustaqil storage hai - ghumti hui plates par data likha jata hai. Purani **Floppy Disk** sirf 1.44 MB rakhti thi. **Magnetic Tape** bare backup ke liye istemal hoti thi.
- **Storage Devices - Optical:** ye laser se parhi/likhi jati hain. **CD** ki capacity lag bhag **700 MB**, **DVD** ki **4.7 GB**, aur **Blu-ray Disc** ki **25 GB** hoti hai. Ye teen numbers test ke pasandeeda sawal hain - teeno pakke yaad karein.
- **Solid State Storage: SSD (Solid State Drive)** mein koi ghumta hua hissa nahi hota, is liye ye HDD se bohat **tez, khamosh aur halka** hai - magar mehnga. USB flash drive (pen drive) aur memory card bhi isi flash memory technology par chalte hain. Aaj kal ke naye laptops mein SSD hi aata hai.
- **Input + Output dono: Touch Screen** ek aisi device hai jo input bhi hai aur output bhi - hum ungli se chhoo kar hidayat dete hain (input) aur screen par nateeja dekhte hain (output). Isi tarah **Modem** bhi data bhejta bhi hai aur leta bhi hai.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Optical discs ki capacity - test wale numbers

CD: lag bhag 700 MB.

DVD (single layer): 4.7 GB.

Blu-ray (single layer): 25 GB.

Floppy disk: sirf 1.44 MB (ab istemal nahi hoti).

Box 2: Khaas devices aur un ke kaam

OMR: MCQ answer sheets check karna.

MICR: bank cheques parhna.

OCR: photo/scan ki likhai ko text banana.

Plotter: bare naqshay aur blueprints print karna.

Aam Ghaltiyan - Common Mistakes

- Printer ki SPEED PPM mein aur QUALITY (resolution) DPI mein hoti hai - dono ko aapas mein badal kar sawal aata hai.
- Scanner input device hai, printer output - naye students in dono ko aksar ulta kar dete hain.
- Touch screen sirf input nahi - woh input AUR output dono hai.
- CD 700 MB, DVD 4.7 GB, Blu-ray 25 GB - numbers ko aapas mein ulta karna aam ghalti hai.

Practice MCQs

1. Which of the following is an input device?

(a) Monitor (b) Scanner (c) Printer (d) Speaker

2. MICR is mostly used in which field?

(a) Banks, for processing cheques (b) Hospitals (c) Schools (d) Airports

3. The device used to check MCQ answer sheets by reading dark marks is:

(a) OCR (b) OMR (c) MICR (d) Barcode reader

4. Which of the following is an output device?

(a) Printer (b) Keyboard (c) Mouse (d) Scanner

5. Which of the following is an impact printer?

(a) Inkjet printer (b) Laser printer (c) Thermal printer (d) Dot Matrix printer

6. The speed of a printer is measured in:

(a) DPI (b) GHz (c) PPM (Pages Per Minute) (d) MB

7. The resolution (print quality) of a printer is measured in:

(a) DPI (Dots Per Inch) (b) PPM (c) bps (d) MHz

8. The storage capacity of a standard CD is about:

(a) 4.7 GB (b) 25 GB (c) 1.44 MB (d) 700 MB

9. The storage capacity of a single-layer DVD is about:

(a) 700 MB (b) 25 GB (c) 4.7 GB (d) 100 GB

10. The storage capacity of a single-layer Blu-ray disc is about:

(a) 4.7 GB (b) 25 GB (c) 700 MB (d) 1 TB

11. Which storage device has no moving parts and works much faster than a hard disk?

(a) HDD (b) Floppy disk (c) SSD (Solid State Drive) (d) CD-ROM

12. A touch screen is an example of:

(a) Only input device (b) Both input and output device (c) Only output device (d) Storage device

13. The smallest element (dot) of a picture on a computer screen is called:

(a) Pixel (b) Byte (c) Dot matrix (d) Cursor

14. Which output device is used to print large maps and engineering drawings?

(a) Inkjet printer (b) Scanner (c) Monitor (d) Plotter

15. A barcode reader is an example of:

(a) Input device (b) Output device (c) Storage device (d) Processing device

16. Which of the following is a secondary storage device?

(a) RAM (b) Hard disk (c) ROM (d) Cache memory

17. A USB flash (pen) drive uses which type of memory?

(a) Magnetic memory (b) Optical memory (c) Flash memory (d) Punch card memory

Answer Key - Chapter 3

1. (b) 2. (a) 3. (b) 4. (a) 5. (d) 6. (c) 7. (a) 8. (d) 9. (c) 10. (b) 11. (c) 12. (b) 13. (a) 14. (d) 15. (a) 16. (b) 17. (c)

Try Yourself - Quick Drills

1. OMR kis kaam ke liye hota hai?
2. Printer ki quality kis mein napi jati hai?
3. SSD HDD se behtar kyun hai?
4. Bare naqshay print karne wali device kon si hai?
5. Ek CD ki capacity kitni hoti hai?

Drill Answers: 1. MCQ answer sheets check karne ke liye 2. DPI (Dots Per Inch) mein 3. Us mein moving parts nahi, woh tez aur khamosh hai 4. Plotter 5. Lag bhag 700 MB

Chapter 4

Software aur Operating System

Hardware woh jisam hai jo dikhta hai, software us ki rooh hai jo kaam karwata hai. System software, application software, compiler-interpreter ka farq aur operating system ke kaam - ye sub is chapter mein hai.

Notes - Asaan Alfaaz Mein

- **Software kya hai:** computer ko diye jane wale **programs (instructions ka majmua)** ko software kehte hain. Hardware ko hum chhoo sakte hain, software ko nahi. Bina software ke computer ek bekaar dabba hai - software hi batata hai ke hardware kya kare.
- **System Software:** ye woh programs hain jo **khud computer ko chalate** hain aur hardware ko control karte hain. Sab se aham system software **Operating System (OS)** hai - jese Windows, Linux, macOS, Android. Is ke ilawa language translators (compiler, interpreter) aur utility programs bhi system software mein aate hain.
- **Application Software:** ye **user ke khaas kaamon** ke liye bane hote hain - MS Word (likhna), MS Excel (hisaab), Chrome (internet), Photoshop (tasveer), VLC (video). Inhein aksar "apps" kaha jata hai. Ye OS ke baghair nahi chal sakte - pehle OS chalta hai, phir applications.
- **Operating System ke kaam:** (1) computer ko start karna (**booting**); (2) files aur folders ko sambhalna (file management); (3) memory (RAM) ko programs mein bantna; (4) input/output devices ko control karna; (5) user ko screen par aasan mahol (interface) dena; (6) ek saath kai programs chalana (multitasking).
- **GUI aur CUI: GUI (Graphical User Interface)** mein tasveeron, icons aur mouse se kaam hota hai - Windows aur Android GUI hain, is liye har koi aasani se seekh leta hai. **CUI/CLI (Command Line Interface)** mein commands type karni padti hain - purana MS-DOS aur Linux ki terminal is ki misal hai. Test mein "Windows kis interface ki misal hai" ka jawab GUI hai.
- **Compiler aur Interpreter:** computer sirf machine language (0 aur 1) samajhta hai, is liye high-level language ke program ko translate karna padta hai. **Compiler** poore program ko **ek hi dafa** translate karta hai aur ghaltiyan ek saath batata hai. **Interpreter** program ko **ek ek line** karke translate aur chalata hai - ghalat line par foran ruk jata hai. **Assembler** assembly language ko machine language mein badalta hai.
- **Utility Software:** ye computer ki dekh bhaal aur safai ke chhote programs hain - **Antivirus** (virus se hifazat), file compression (WinRAR - file ka size chhota karna), backup programs, disk cleanup. Ye system software ki ek qisam mana jata hai.
- **Firmware aur Open Source:** jo software hardware chip ke andar mustaqil tor par dala jaye, use **Firmware** kehte hain - BIOS is ki misal hai. **Open source** software ka code sab ke liye khulla hota hai aur woh aksar muft hota hai - Linux is ki sab se mashhoor misal. Windows proprietary (paid) software hai.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Software ki aqsaam - ek nazar mein

System software: OS (Windows, Linux), compiler, antivirus utilities.

Application software: MS Word, Excel, Chrome, Photoshop.

Firmware: chip mein mustaqil software - BIOS.

Yaad rakhein: pehle OS, phir applications chalti hain.

Box 2: Compiler aur Interpreter - farq

Compiler: poora program ek saath translate, ghaltiyan list mein.

Interpreter: line by line translate, ghalat line par foran ruk jaye.

Assembler: assembly language ko machine code mein badalta hai.

Teeno language translators (system software) hain.

Aam Ghaltiyan - Common Mistakes

- Antivirus application nahi, UTILITY (system) software hai - ye sawal aksar aata hai.
- Compiler poora program ek saath, interpreter line by line - ye farq ulta hone par number katta hai.
- Linux open source aur muft hai, Windows proprietary - dono OS hain magar licence alag hai.
- Booting ka matlab computer ko on karke OS ko RAM mein load karna hai, koi program kholna nahi.

Practice MCQs

1. A set of instructions or programs that tells the computer what to do is called:

(a) Hardware (b) Firmware device (c) Input (d) Software

2. The most important system software of a computer is:

(a) MS Word (b) Web browser (c) Operating System (d) Antivirus only

3. Which of the following is an operating system?

(a) Linux (b) MS Excel (c) Google Chrome (d) Photoshop

4. A compiler translates the program:

(a) Line by line (b) Word by word on paper (c) Only when printed (d) All at once (whole program in one go)

5. An interpreter translates and executes a program:

(a) All at once (b) Line by line (c) Only one time in a year (d) Without running it

6. The translator that converts assembly language into machine language is called:

(a) Compiler (b) Assembler (c) Interpreter (d) Browser

7. Antivirus is an example of:

(a) Application software (b) Graphics software (c) Word processor (d) Utility software

8. Windows uses which type of user interface?

(a) CUI (Command Line Interface) (b) Voice only interface (c) GUI (Graphical User Interface) (d) Punch card interface

9. MS-DOS is an example of:

(a) CUI / Command Line Interface (b) GUI (c) Touch interface (d) Web interface

10. Software permanently stored inside a hardware chip, such as BIOS, is called:

(a) Application software (b) Freeware (c) Malware (d) Firmware

11. Which of the following is application software?

(a) Windows (b) MS Excel (c) Linux (d) BIOS

12. Linux is best known as:

(a) Open source operating system (b) Paid word processor (c) Antivirus program (d) Video game only

13. Loading the operating system into RAM when the computer is switched on is called:

(a) Formatting (b) Scanning (c) Booting (d) Downloading

14. Files, folders and memory of a computer are mainly managed by:

(a) Operating System (b) Printer (c) MS Word (d) Monitor

15. The only language which a computer can directly understand is:

(a) English language (b) BASIC language (c) Machine language (d) Urdu language

16. WinRAR is used for:

(a) Playing songs (b) Compressing files to reduce their size (c) Making slides (d) Removing the CPU

17. GUI stands for:

(a) General User Internet (b) Graphical Unit Input (c) Global User Interface (d) Graphical User Interface

Answer Key - Chapter 4

1. (d) 2. (c) 3. (a) 4. (d) 5. (b) 6. (b) 7. (d) 8. (c) 9. (a) 10. (d) 11. (b) 12. (a) 13. (c) 14. (a) 15. (c) 16. (b) 17. (d)

Try Yourself - Quick Drills

1. Operating system ka sab se aham kaam kya hai?
2. Compiler aur interpreter mein kya farq hai?
3. GUI ka full form kya hai?
4. Firmware ki ek misal dein.
5. Open source OS ki misal dein.

Drill Answers: 1. Poore computer ko chalana aur tamam hisso ko manage karna 2. Compiler poora program ek saath, interpreter line by line translate karta hai 3. Graphical User Interface 4. BIOS 5. Linux

Chapter 5

MS Word - Word Processing

MS Word duniya ka sab se zyada istemal hone wala word processor hai. Uski file extension, default font, mail merge aur shortcut keys har office aur teacher test mein aati hain.

Notes - Asaan Alfaaz Mein

- **MS Word kya hai:** ye Microsoft ka **word processing** software hai - letters, applications, reports aur documents banane ke liye. Word document ki file extension naye versions mein **.docx** hai (purane Word 2003 tak **.doc** thi). Ye farq test ka pasandeeda sawal hai.
- **Default settings:** naye MS Word mein default font **Calibri** aur default font size **11 points** hota hai. Default page orientation **Portrait** (khari/vertical) hai; chaure (horizontal) page ko **Landscape** kehte hain. Page ke chaaron kinare ke khaali hisse ko **Margin** kehte hain.
- **Header aur Footer:** har page ke **upar** dohraaya jane wala text (jese school ka naam) **Header** mein aur har page ke **neeche** wala text (jese page number) **Footer** mein likha jata hai. Ek dafa likhne se tamam pages par aa jata hai.
- **Mail Merge:** ye Word ki sab se taqatwar khubi hai - ek hi letter ko **bohat se logon ke naam aur address** ke saath alag alag print karna. School ke result cards aur dawat-naame is se bante hain. Is mein ek main document aur ek data list (naam/address wali) hoti hai.
- **Spelling aur Grammar check:** Word ghalat spelling ke neeche **surkh (red)** line aur grammar ki ghalti ke neeche rang-daar line dikhata hai. Spelling check chalane ki key **F7** hai, aur synonyms (ham-maani alfaaz) ke liye **Thesaurus** istemal hota hai - us ki shortcut Shift+F7 hai.
- **Aham Shortcuts - Save/Print/Edit: Ctrl+S** save, **Ctrl+P** print, **Ctrl+O** open, **Ctrl+N** new document, **Ctrl+X** cut, **Ctrl+C** copy, **Ctrl+V** paste, **Ctrl+Z** undo (wapas), **Ctrl+Y** redo. Ye shortcuts tamam MS Office programs mein ek jese chalte hain.
- **Aham Shortcuts - Dhoondhna aur Formatting: Ctrl+F** find (lafz dhoondhna), **Ctrl+H** replace (lafz badalna), **Ctrl+B** bold, **Ctrl+I** italic, **Ctrl+U** underline, **Ctrl+L/E/R/J** alignment (left, center, right, justify), **Ctrl+A** select all. **Superscript** (x2 jesa upar wala) Ctrl+Shift++ aur **Subscript** (H2O jesa neeche wala) Ctrl+= se banta hai.
- **Word ki aur khaas baatein:** Word mein zoom zyada se zyada **500%** tak ho sakta hai. Sirf line badalne (bina naya paragraph kiye) ke liye **Shift+Enter** dabate hain. Ek line ke aakhir par lafz khud agli line par chala jana **Word Wrap** kehlata hai. Document dekhne ka default view **Print Layout** hai.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: MS Word shortcuts - lazmi yaad karein

Ctrl+S: Save. Ctrl+P: Print. Ctrl+N: New. Ctrl+O: Open.

Ctrl+X/C/V: Cut, Copy, Paste. Ctrl+Z: Undo. Ctrl+Y: Redo.

Ctrl+F: Find. Ctrl+H: Replace. Ctrl+A: Select All.

F7: Spelling check. Shift+F7: Thesaurus.

Box 2: Header, Footer aur Mail Merge kwik yaad-dehani

Header: har page ke upar wala repeated text.

Footer: har page ke neeche wala text - page number yahin ata hai.

Mail Merge: ek letter, bohat se naam-address ke saath print.

Watermark: page ke peeche halka sa text (jese "CONFIDENTIAL").

Aam Ghaltiyan - Common Mistakes

- Find Ctrl+F hai aur Replace Ctrl+H - dono ko aapas mein badalna sab se aam ghalti hai.
- Neeche wala hissa FOOTER hai, upar wala HEADER - naam ulta lagana ghalat hai.
- .docx naya extension hai, .doc purana (2003 tak) - version poocha jaye to ghor karein.
- Spelling check F7 hai; F5 Word mein Go To ke liye hai, slide show ke liye nahi (woh PowerPoint hai).

Practice MCQs

1. The default file extension of a document in new versions of MS Word is:

(a) .doc (b) .xlsx (c) .pptx (d) .docx

2. The shortcut key to save a document is:

(a) Ctrl + S (b) Ctrl + P (c) Ctrl + N (d) Ctrl + O

3. The shortcut key for spelling and grammar check in MS Word is:

(a) F5 (b) F1 (c) F7 (d) F12

4. The Thesaurus in MS Word is opened by:

(a) Ctrl + F7 (b) Shift + F7 (c) Alt + F7 (d) F9

5. The shortcut for Find and Replace in MS Word is:

(a) Ctrl + F (b) Ctrl + H (c) Ctrl + R (d) Ctrl + G

6. The default font in new versions of MS Word is:

(a) Times New Roman (b) Arial (c) Calibri (d) Courier New

7. The default font size in new versions of MS Word is:

(a) 10 points (b) 12 points (c) 14 points (d) 11 points

8. The default page orientation in MS Word is:

(a) Portrait (b) Landscape (c) Square (d) Book fold

9. The feature used to send the same letter to many persons with their own names and addresses is:

(a) Word Wrap (b) AutoCorrect (c) Mail Merge (d) Track Changes

10. The text that appears at the top of every page of a document is written in the:

(a) Footer (b) Margin (c) Footnote (d) Header

11. The shortcut key for superscript (writing a small character above the line) is:

(a) Ctrl + Shift + + (b) Ctrl + = (c) Ctrl + B (d) Alt + Shift

12. The maximum zoom percentage in MS Word is:

(a) 200% (b) 500% (c) 300% (d) 1000%

13. The shortcut key for Undo is:

(a) Ctrl + Y (b) Ctrl + X (c) Ctrl + U (d) Ctrl + Z

14. The shortcut key for Redo is:

(a) Ctrl + Z (b) Ctrl + R (c) Ctrl + Y (d) Ctrl + D

15. The default view in which a document is shown in MS Word is:

(a) Web Layout (b) Print Layout (c) Outline view (d) Draft view only

16. To move to a new line without starting a new paragraph, we press:

(a) Shift + Enter (b) Ctrl + Enter (c) Alt + Enter (d) Tab only

17. The automatic shifting of a word to the next line when it does not fit is called:

(a) Mail Merge (b) Word Wrap (c) Word Count (d) Text Box

Answer Key - Chapter 5

1. (d) 2. (a) 3. (c) 4. (b) 5. (b) 6. (c) 7. (d) 8. (a) 9. (c) 10. (d) 11. (a) 12. (b) 13. (d) 14. (c) 15. (b) 16. (a) 17. (b)

Try Yourself - Quick Drills

1. Mail Merge kis kaam aata hai?
2. Header aur Footer mein kya farq hai?
3. Spelling check ki shortcut key kya hai?
4. MS Word document ki default extension kya hai?
5. Portrait aur Landscape mein kya farq hai?

Drill Answers: 1. Ek hi letter bohat se logon ko un ke naam-address ke saath bhejna/print karna 2. Header page ke upar, Footer page ke neeche hota hai 3. F7 4. .docx 5. Portrait khara (vertical) page, Landscape chaura (horizontal) page

Chapter 6

MS Excel aur MS PowerPoint

Excel hisaab-kitab aur data ke liye, PowerPoint presentations ke liye. Workbook, cell, formula-functions aur slide show shortcuts - is chapter ke har sawal ka test mein numaya chance hai.

Notes - Asaan Alfaaz Mein

- **MS Excel kya hai:** ye **spreadsheet** software hai - rows aur columns ki grid mein data rakh kar hisaab karne ke liye. Excel ki file ko **Workbook** kehte hain, us ke andar wale pages ko **Worksheet** kehte hain. File extension **.xlsx** hai.
- **Cell aur Cell Address:** row aur column ke milne ki jagah ko **Cell** kehte hain - har data aik cell mein rakha jata hai. Har cell ka naam/patta us ke column letter aur row number se banta hai - jese **A1** (pehla column, pehli row). Excel 2016 ki ek worksheet mein **1,048,576 rows** aur 16,384 columns (aakhri column XFD) hote hain. Naye workbook mein default tor par 1 worksheet hoti hai.
- **Formula:** Excel mein har hisaab **= (barabar ke nishan)** se shuru hota hai. Agar = na lagayen to Excel use aam text samjhega aur hisaab nahi karega. Misal: =A1+B1 likhne se do cells ki qeematein jama ho jati hain.
- **Aham Functions: SUM** numbers ko jama karta hai, **AVERAGE** ausat nikalta hai, **MAX** sab se bari qeemat, **MIN** sab se chhoti qeemat, **COUNT** numbers wale cells ginta hai, **IF** shart (condition) par faisla karta hai. Functions formula ke andar likhe jate hain - jese =SUM(A1:A5).
- **Charts:** Excel data ko tasveeri shakal (graph) mein dikhata hai - column chart, pie chart, line chart waghera. Chart se data samajhna asan ho jata hai aur ye bhi ek aam test sawal hai.
- **MS PowerPoint kya hai:** ye **presentation** banane ka software hai. Is ki file extension **.pptx** hai. Presentation ke har page ko **Slide** kehte hain. Pehli slide aam tor par **Title Slide** layout ki hoti hai - us par topic aur pesh karne wale ka naam hota hai.
- **Slide Show Shortcuts:** slide show **shuru se** chalane ke liye **F5** dabate hain, aur **jo slide khuli ho wahan se** chalane ke liye **Shift+F5**. Show khatam karne ke liye Esc dabate hain. Ye dono shortcuts test mein bar bar aate hain.
- **Transition aur Animation ka farq: Transition** woh effect hai jo **ek slide se dusri slide** par jane mein lagta hai. **Animation** woh harkat hai jo **ek hi slide ke andar** kisi text ya tasveer par lagti hai. Dono ka farq test ka pasandeeda sawal hai. Audience ko dene wali slides ki print shakal ko **Handouts** kehte hain.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Excel functions - fori guide

SUM: jama karna - =SUM(A1:A5).

AVERAGE: ausat - =AVERAGE(A1:A5).

MAX / MIN: sab se bari / sab se chhoti qeemat.

COUNT: numbers wale cells ginna. IF: shart par faisla.

Box 2: PowerPoint - transition vs animation

Transition: slides ke darmiyan ka effect (dusri slide kese aaye).

Animation: slide ke andar text/tasveer ki harkat.

F5: show shuru se. Shift+F5: current slide se.

Handouts: audience ke liye slides ka print.

Aam Ghaltiyan - Common Mistakes

- Formula hamesha = se shuru hota hai - baghair = ke Excel hisaab nahi karega.
- Transition slides ke darmiyan hota hai, animation slide ke andar - ye farq kabhi na bhoolein.
- Workbook poori file hai, worksheet us ka ek page - file aur page ko ulta karna ghalat hai.
- Excel ki extension .xlsx aur PowerPoint ki .pptx hai - Word wali .docx se mix na karein.

Practice MCQs

1. An MS Excel file is called a:

(a) Workbook (b) Worksheet (c) Document (d) Slide book

2. The intersection of a row and a column in Excel is called a:

(a) Field (b) Range box (c) Table (d) Cell

3. Every formula in MS Excel begins with the sign:

(a) + (plus) (b) # (hash) (c) = (equal) (d) @ (at)

4. Which function is used to add (sum) values in Excel?

(a) SUM (b) ADD (c) TOTAL (d) PLUS

5. Which function is used to find the average of values in Excel?

(a) AVG-TOTAL (b) MEAN-IT (c) SUM (d) AVERAGE

6. The file extension of an MS Excel workbook is:

(a) .docx (b) .pptx (c) .xlsx (d) .exlx

7. The address of the very first cell of a worksheet is:

(a) 1A (b) A1 (c) AA (d) B1

8. How many rows are there in one worksheet of Excel 2016?

(a) 65,536 (b) 1,00,000 (c) 1,048,576 (d) 16,384

9. Which function counts the number of cells containing numbers?

(a) SUM (b) COUNT (c) MAX (d) TOTAL

10. By default, a new workbook in Excel 2016 contains how many worksheets?

(a) 1 (b) 3 (c) 5 (d) 10

11. The file extension of an MS PowerPoint presentation is:

(a) .docx (b) .xlsx (c) .ptpx (d) .pptx

12. The shortcut key to start the slide show from the beginning is:

(a) F5 (b) F7 (c) Shift + F5 (d) Ctrl + F5

13. The shortcut key to start the slide show from the current slide is:

(a) F5 (b) Shift + F5 (c) F2 (d) Ctrl + Shift + S

14. The special effects that appear when moving from one slide to the next are called:

(a) Animations (b) Slide layouts (c) Transitions (d) Handouts

15. The movement effects applied to text or objects within a single slide are called:

(a) Transitions (b) Slide shows (c) Themes only (d) Animations

16. Printed copies of slides given to the audience are called:

(a) Outlines (b) Notes pages (c) Handouts (d) Slide sorters

17. The layout normally used for the very first slide of a presentation is:

(a) Title Slide (b) Blank layout (c) Two Content layout (d) Table layout

Answer Key - Chapter 6

1. (a) 2. (d) 3. (c) 4. (a) 5. (d) 6. (c) 7. (b) 8. (c) 9. (b) 10. (a) 11. (d) 12. (a) 13. (b) 14. (c) 15. (d) 16. (c) 17. (a)

Try Yourself - Quick Drills

1. Excel mein cell kise kehte hain?
2. Numbers jama karne ka function kon sa hai?
3. Slide show shuru se chalane ki key kya hai?
4. Excel file ki extension kya hai?
5. Transition aur Animation mein kya farq hai?

Drill Answers: 1. Row aur column ke milne ki jagah ko 2. SUM 3. F5 4. .xlsx 5. Transition slides ke darmiyan, animation slide ke andar lagta hai

Chapter 7

Internet aur Networking

Internet aaj har test ka lazmi hissa hai. Browser aur search engine ka farq, LAN-WAN, modem-router, DNS aur protocols (HTTP, FTP, SMTP) - is chapter ke aham tareen topics hain.

Notes - Asaan Alfaaz Mein

- **Internet kya hai:** internet duniya bhar ke computers ka **sab se bara network** hai - lakhoon chhote networks ka jura hua majmua. Ye kisi ek shakhs ya idare ki milkiyat nahi. Is par hum websites dekhte hain, email bhejte hain, files share karte hain aur video dekhte hain.
- **WWW (World Wide Web):** WWW internet par maujood **web pages ka majmua** hai. Ise **Tim Berners-Lee** ne 1989-91 mein ijaad kiya. Ek baat pakki rakhein: **WWW aur Internet ek cheez nahi** - internet bunyaad (raasta) hai, WWW us par chalne wali ek service hai. Email bhi internet ki service hai magar WWW nahi.
- **Browser aur Search Engine ka farq: Web browser** woh program hai jis se hum websites kholte hain - **Google Chrome, Mozilla Firefox, Microsoft Edge**. **Search engine** woh website hai jo internet par cheezein dhoondhti hai - **Google, Bing, Yahoo**. Chrome browser hai, Google search engine - test mein yahi farq phasata hai.
- **Website, Webpage, Homepage:** jure hue web pages ke majmue ko **Website** kehte hain, us ka har ek page **Webpage** hai, aur website ka **pehla page Homepage** kehlata hai. Jis computer par websites rakhi hoti hain use **Web Server** kehte hain.
- **URL aur Protocols:** har webpage ka apna makhsoos patta **URL (Uniform Resource Locator)** hota hai - jese https://www.example.com. Web pages lane ka usool (protocol) **HTTP** hai; us ki mehfooz shakal **HTTPS** hai jahan data lock (encrypted) ho kar jata hai - bank aur shopping websites par HTTPS zaroor dekhein (address bar mein tala/lock ka nishan).
- **Aur aham protocols: FTP (File Transfer Protocol)** files ko ek computer se dusre par bhejne ke liye; **SMTP** email bhejne ke liye; **TCP/IP** internet ka bunyadi usool hai jo data ko chhote packets mein tor kar sahi jagah pohchata hai. Har protocol ek khaas kaam ke usoolon ka naam hai.
- **Networks ki aqsaam (LAN, MAN, WAN): LAN (Local Area Network)** chhoti jagah ka network hai - ek school, office ya ghar. **MAN (Metropolitan Area Network)** ek sheher tak phaila hota hai - cable TV network ki misal. **WAN (Wide Area Network)** mulkon aur barron tak phaila hota hai - internet khud sab se bara WAN hai.
- **Network devices: Modem** computer ke digital signal ko telephone line ke signal mein aur wapas badalta hai - naam hi Modulator-Demodulator se bana hai. **Router** data ko sahi raaste se sahi network tak pohchata hai - ghar ke Wi-Fi router se kai devices ek saath internet use karte hain. **Switch/Hub** ek hi network ke andar computers ko jorte hain.
- **DNS aur IP Address:** har computer ka internet par ek number wala patta hota hai - **IP Address** (jese 192.168.1.1). Numbers yaad rakhna mushkil hai, is liye hum naam (domain name) likhte hain - **DNS (Domain Name System)** us naam ko IP address mein badal deta hai. DNS ko internet ki phone book kaha jata hai. **Wi-Fi** baghair taar (wireless) ke LAN jorne ki technology hai. Data transfer ki raftaar **bps (bits per second)** mein napi jati hai.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Browser vs Search Engine - ulta na karein

Browser (websites kholne wala program): Chrome, Firefox, Edge.

Search Engine (dhoondhne wali website): Google, Bing, Yahoo.

Google search engine hai; Chrome Google ka browser hai.

Website server par hoti hai, browser use dikhata hai.

Box 2: Network ki aqsaam - size ke lehaz se

LAN: ek building - school, office, ghar.

MAN: ek sheher - cable TV network.

WAN: mulk se mulk tak - internet sab se bara WAN.

Har network size mein barhta jata hai: LAN, MAN, WAN.

Aam Ghaltiyan - Common Mistakes

- Google search engine hai, browser NAHI - browser Chrome hai. Ye test ki sab se mashhoor ghalti hai.
- Internet aur WWW ek cheez nahi - WWW internet ki sirf ek service hai.
- Email bhejne ka protocol SMTP hai, lane/transfer ka FTP - dono ke kaam mix na karein.
- LAN sab se chhota network hai, WAN sab se bara - "internet LAN hai" wala option hamesha ghalat hota hai.

Practice MCQs

1. The World Wide Web (WWW) was invented by:

(a) Bill Gates (b) Charles Babbage (c) Steve Jobs (d) Tim Berners-Lee

2. WWW stands for:

(a) World Web Window (b) World Wide Web (c) Wide World Web (d) Web World Wide

3. Which protocol is used for secure (locked/encrypted) browsing of websites?

(a) HTTP (b) HTTPS (c) FTP (d) SMTP

4. Which of the following is a web browser?

(a) Google (b) Bing (c) Yahoo (d) Google Chrome

5. A network spread over a small area such as one school or office is called:

(a) MAN (b) WAN (c) LAN (d) Internet

6. The largest example of a WAN is:

(a) Internet (b) A school LAN (c) A home Wi-Fi (d) Bluetooth

7. The device which converts digital signals of a computer into telephone-line signals and back is:

(a) Printer (b) Scanner (c) Monitor (d) Modem

8. The main job of DNS is to change:

(a) Images into text (b) Domain names into IP addresses (c) Files into folders (d) Emails into letters

9. The unique address of a webpage on the internet is called:

(a) URL (b) Email (c) IP of user (d) Domain password

10. URL stands for:

(a) Universal Register List (b) Uniform Resource Letter (c) Uniform Resource Locator (d) United Resource Locator

11. Which protocol is mainly used for sending e-mail?

(a) SMTP (b) FTP (c) HTTP (d) POP server only

12. Which protocol is used to transfer files between computers?

(a) SMTP (b) DNS (c) FTP (d) HTML

13. In a star topology, all computers are connected to:

(a) Each other in a circle (b) A central device (hub/switch) (c) One long single cable only (d) The printer

14. The speed of data transfer on a network is measured in:

(a) DPI (b) PPM (c) GHz only (d) bps (bits per second)

15. IP in "IP Address" stands for:

(a) Internal Program (b) Internet Page (c) Input Protocol (d) Internet Protocol

16. The computer on which websites are stored and from which they are served is called:

(a) Web Server (b) Web Browser (c) Search Engine (d) Modem

17. The first page of a website is called its:

(a) Login page (b) Browser page (c) Home page (d) Server page

Answer Key - Chapter 7

1. (d) 2. (b) 3. (b) 4. (d) 5. (c) 6. (a) 7. (d) 8. (b) 9. (a) 10. (c) 11. (a) 12. (c) 13. (b) 14. (d) 15. (d) 16. (a) 17. (c)

Try Yourself - Quick Drills

1. LAN aur WAN mein kya farq hai?
2. Modem kya kaam karta hai?
3. DNS ka kaam kya hai?
4. Browser aur search engine mein kya farq hai?
5. HTTPS mein "S" ka kya matlab hai?

Drill Answers: 1. LAN chhoti jagah (building) ka, WAN bare ilaqe (mulk-duniya) ka network hai 2. Digital aur line signal ko ek dusre mein badalta hai taake internet jud sake 3. Website ke naam ko IP address mein badalna 4. Browser websites kholta hai, search engine cheezein dhoondhta hai 5. Secure - data lock (encrypted) ho kar jata hai

Chapter 8

Abbreviations - Full Forms

Full forms computer test ka sab se asan number dene wala hissa hai. Is chapter mein hardware, memory, internet aur files se judi tamam mashhoor abbreviations jama kar di gayi hain.

Notes - Asaan Alfaaz Mein

- **Full forms kyun aham hain:** har PST/JEST aur screening test mein 3-5 sawal seedha full forms par hote hain. Inhein groups mein yaad karein - hardware wale, internet wale, file wale - to bhoolna mushkil ho jata hai. Neeche tamam mashhoor full forms de di gayi hain; MCQs mein inhi ki practice hai.
- **Hardware aur Memory group: CPU** = Central Processing Unit, **ALU** = Arithmetic Logic Unit, **RAM** = Random Access Memory, **ROM** = Read Only Memory, **USB** = Universal Serial Bus, **HDMI** = High Definition Multimedia Interface, **SSD** = Solid State Drive, **HDD** = Hard Disk Drive, **LCD** = Liquid Crystal Display, **LED** = Light Emitting Diode.
- **Internet group: WWW** = World Wide Web, **HTTP** = HyperText Transfer Protocol, **URL** = Uniform Resource Locator, **IP** = Internet Protocol, **DNS** = Domain Name System, **ISP** = Internet Service Provider (jo company internet deti hai), **Wi-Fi** aam tor par Wireless Fidelity se jura samjha jata hai, **FTP** = File Transfer Protocol, **SMTP** = Simple Mail Transfer Protocol.
- **Files aur Media group: PDF** = Portable Document Format, **JPEG** = Joint Photographic Experts Group (tasveer wali file), **MP3** audio file ka mashhoor format hai, **CD** = Compact Disc, **DVD** = Digital Versatile Disc, **GPS** = Global Positioning System (jagah maloom karne wala nizaam).
- **General group: ICT** = Information and Communication Technology, **IT** = Information Technology, **AI** = Artificial Intelligence, **VGA** = Video Graphics Array (purani display technology), **SIM** = Subscriber Identity Module (mobile ki chip), **ATM** = Automated Teller Machine, **PIN** = Personal Identification Number.
- **Yaad karne ki trick:** jo full form mushkil lage, use us ke kaam se jor kar yaad karein - ISP "internet dene wala", GPS "jagah batane wala", PDF "har jagah khulne wali file". Test mein ghalat options aksar ek lafz badal kar bane hote hain - jese "Universal Serial Bus" ki jagah "United Serial Bus" - is liye har lafz ghor se parhein.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Hardware group - ek nazar mein

CPU: Central Processing Unit. ALU: Arithmetic Logic Unit.

RAM: Random Access Memory. ROM: Read Only Memory.

USB: Universal Serial Bus. HDMI: High Definition Multimedia Interface.

LCD: Liquid Crystal Display. LED: Light Emitting Diode.

Box 2: Internet group - ek nazar mein

HTTP: HyperText Transfer Protocol. URL: Uniform Resource Locator.

IP: Internet Protocol. DNS: Domain Name System.

ISP: Internet Service Provider. FTP: File Transfer Protocol.

SMTP: Simple Mail Transfer Protocol.

Aam Ghaltiyan - Common Mistakes

- USB mein "Serial" hai, "Service" nahi - ek lafz badla hua option test setter ki chaal hai.

- PDF ka "Portable" hai, "Personal" ya "Public" nahi - P se shuru hone wale teenon options ghor se parhein.
- RAM "Random Access" hai aur ROM "Read Only" - dono mein R wale alfaaz aapas mein badalna aam ghalti hai.
- JPEG ka J "Joint" hai - tasveer wali file ko "Jpeg" kehna aur full form bhoolna number kata sakta hai.

Practice MCQs

1. CPU stands for:

(a) Central Program Unit (b) Central Processing Unit (c) Computer Personal Unit (d) Central Processing Utility

2. USB stands for:

(a) United Serial Bus (b) Universal Serial Bus (c) Universal Service Board (d) Universal System Backup

3. HDMI stands for:

(a) High Data Multimedia Input (b) High Definition Media Internet (c) High Definition Multimedia Interface (d) Home Digital Multimedia Interface

4. ISP stands for:

(a) Internet System Protocol (b) International Service Program (c) Internet Server Page (d) Internet Service Provider

5. PDF stands for:

(a) Portable Document Format (b) Personal Document File (c) Public Data Format (d) Printed Document Form

6. JPEG stands for:

(a) Joint Picture Editing Group (b) Japan Photographic Experts Group (c) Joint Photographic Experts Group (d) Joint Photographic Editing Graphics

7. GPS stands for:

(a) General Positioning Service (b) Global Program Satellite (c) Geographic Phone System (d) Global Positioning System

8. LCD stands for:

(a) Liquid Crystal Display (b) Light Crystal Display (c) Liquid Colour Display (d) Laser Crystal Device

9. RAM stands for:

(a) Read And Memory (b) Random Access Memory (c) Rapid Access Module (d) Random Active Memory

10. ROM stands for:

(a) Random Only Memory (b) Read Open Memory (c) Rapid Operating Memory (d) Read Only Memory

11. HTTP stands for:

(a) HyperText Transmission Program (b) High Text Transfer Protocol (c) HyperText Transfer Protocol (d) HyperText Transfer Page

12. ICT stands for:

(a) Internet and Computer Technology (b) Information and Communication Technology (c) Information and Computing Tools (d) International Communication Technology

13. CD stands for:

(a) Compact Disc (b) Computer Disc (c) Compact Data (d) Central Disc

14. DVD stands for:

(a) Digital Video Data (b) Digital Versatile Disc (c) Data Versatile Drive (d) Digital Versatile Data

15. AI stands for:

(a) Artificial Intelligence (b) Automatic Intelligence (c) Artificial Internet (d) Advanced Information

16. VGA stands for:

(a) Visual Graphics Adapter (b) Video General Array (c) Virtual Graphics Access (d) Video Graphics Array

17. SIM (in a mobile phone) stands for:

(a) Subscriber Internet Module (b) System Identity Machine (c) Subscriber Identity Module (d) Subscriber Information Memory

Answer Key - Chapter 8

1. (b) 2. (b) 3. (c) 4. (d) 5. (a) 6. (c) 7. (d) 8. (a) 9. (b) 10. (d) 11. (c) 12. (b) 13. (a) 14. (b) 15. (a) 16. (d) 17. (c)

Try Yourself - Quick Drills

1. USB ka full form batayein.

2. PDF ka full form batayein.

3. DNS ka full form batayein.

4. GPS ka full form batayein.

5. ICT ka full form batayein.

Drill Answers: 1. Universal Serial Bus 2. Portable Document Format 3. Domain Name System 4. Global Positioning System 5. Information and Communication Technology

Chapter 9

Number Systems aur Binary

Computer andar se sirf 0 aur 1 samajhta hai. Binary, decimal, octal, hexadecimal ki tabdeeli (conversion) aur ASCII code is chapter ka khulasa hai - thori practice se ye sawal pakke number dete hain.

Notes - Asaan Alfaaz Mein

- **Number System kya hai:** numbers likhne ke usool ko number system kehte hain. Har system ki ek **Base (buniyaad)** hoti hai - jitni base, utne hi alag nishan (digits) istemal hote hain. Rozana zindagi mein hum **Decimal (base 10)** istemal karte hain: digits 0 se 9.
- **Binary (base 2):** binary mein sirf do digits hain - **0 aur 1**. Har binary digit ko **Bit** kehte hain. Computer ki tamam andar ki kaam-kaj binary mein hoti hai kyunke electronic switch ki bhi do hi haalatein hain - ON (1) aur OFF (0).
- **Octal aur Hexadecimal: Octal ki base 8** hai - digits 0 se 7. **Hexadecimal ki base 16** hai - digits 0 se 9 aur haroof **A se F**; is mein A = 10, B = 11, C = 12, D = 13, E = 14, F = 15. Lambi binary ko chhota likhne ke liye ye dono istemal hote hain - 1 hexadecimal digit = 4 binary digits (ek nibble).
- **Binary se Decimal tabdeeli:** har binary digit ki position ki qeemat 2 ki power hoti hai - dayein se bayein: 1, 2, 4, 8, 16, 32... Jahan 1 ho us ki qeemat jama kar dein. Misal: 1010 = 8 + 0 + 2 + 0 = **10**. 1101 = 8 + 4 + 0 + 1 = **13**.
- **Decimal se Binary tabdeeli:** number ko 2 se taqseem karte jayein aur har dafa ka baaqi (remainder) note karein; aakhir mein baaqiyon ko neeche se upar parhein. Misal: 5 ko 2 se taqseem - 5/2 = 2 baaqi 1; 2/2 = 1 baaqi 0; 1/2 = 0 baaqi 1 - neeche se upar: **101**. Isi tarah 10 = 1010 aur 9 = 1001.
- **1 Byte ki taqat:** 1 byte = 8 bits; har bit ki 2 haalatein, to ek byte se **256** mukhtalif qeematein (0 se 255) ban sakti hain. Isi liye ek byte ek harf ya number rakhne ke liye kaafi hota hai.
- **ASCII Code:** computer mein haroof aur nishanaat ko numbers se zahir kiya jata hai - is mashhoor code ka naam **ASCII (American Standard Code for Information Interchange)** hai. Ye 7-bit code hai - kul 128 characters. Yaad rakhein: bare **A ka ASCII code 65** hai, chhote a ka 97, aur digit 0 ka 48. Aaj kal duniya ki tamam zabano ke liye **Unicode** istemal hota hai jo is se bohat bara hai.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Binary se decimal - do misalein

1010: jagah ki qeematein 8, 0, 2, 0 - jama = 10.

1101: 8, 4, 0, 1 - jama = 13.

Usool: dayein se bayein qeematein 1, 2, 4, 8, 16 dugni hoti jati hain.

Sirf wahan jama karein jahan digit 1 ho.

Box 2: Bases ki fori pehchan

Binary: base 2 - digits sirf 0, 1.

Octal: base 8 - digits 0 se 7.

Decimal: base 10 - digits 0 se 9.

Hexadecimal: base 16 - 0 se 9 aur A(10) se F(15).

Aam Ghaltiyan - Common Mistakes

- Hexadecimal mein A ki qeemat 10 hai, 11 nahi - B = 11, C = 12, isi tartib se F = 15.
- Octal mein digit 8 aur 9 NAHI hote - agar number mein 8 dikhe to woh octal ho hi nahi sakta.
- Binary tabdeeli mein baaqi (remainder) ko NEECHE se upar parha jata hai, upar se neeche nahi.
- Computer "decimal" nahi, andar se "binary" samajhta hai - hum decimal likhte hain, machine use binary bana leti hai.

Practice MCQs

1. The base of the binary number system is:

(a) 2 (b) 8 (c) 10 (d) 16

2. The digits 0 and 1 in binary are each called a:

(a) Byte (b) Digit-code (c) Nibble-word (d) Bit

3. The decimal equivalent of binary 1010 is:

(a) 8 (b) 12 (c) 10 (d) 15

4. The binary equivalent of decimal 5 is:

(a) 110 (b) 101 (c) 100 (d) 111

5. In the hexadecimal system, the letters used along with digits are:

(a) A to D (b) G to L (c) A to F (d) Only X and Y

6. The base of the octal number system is:

(a) 2 (b) 8 (c) 10 (d) 16

7. In hexadecimal, the letter A represents the decimal value:

(a) 10 (b) 11 (c) 15 (d) 1

8. The ASCII code of capital letter A is:

(a) 97 (b) 48 (c) 66 (d) 65

9. ASCII stands for:

(a) American Standard Code for Information Interchange (b) American System Code for Internet Information (c) Asian Standard Code for Information Interchange (d) American Standard Computer Information Interchange

10. Inside, a computer works and understands only the _____ number system.

(a) Decimal (b) Binary (c) Octal (d) Hexadecimal

11. The binary equivalent of decimal 10 is:

(a) 1001 (b) 1100 (c) 1010 (d) 1011

12. Which of the following is NOT a number system?

(a) Binary (b) Octal (c) Hexadecimal (d) Alphabetical

13. The total number of different values that 1 byte can represent is:

(a) 8 (b) 100 (c) 256 (d) 1024

14. In hexadecimal, the letter F is equal to decimal:

(a) 15 (b) 10 (c) 16 (d) 12

15. The digits used in the octal number system are:

(a) 0 to 9 (b) 1 to 8 (c) 0 and 1 only (d) 0 to 7

16. The modern code system used to represent characters of almost all languages of the world is:

(a) ASCII only (b) Unicode (c) Binary code (d) Octal code

17. The decimal equivalent of binary 1101 is:

(a) 11 (b) 13 (c) 12 (d) 14

Answer Key - Chapter 9

1. (a) 2. (d) 3. (c) 4. (b) 5. (c) 6. (b) 7. (a) 8. (d) 9. (a) 10. (b) 11. (c) 12. (d) 13. (c) 14. (a) 15. (d) 16. (b) 17. (b)

Try Yourself - Quick Drills

1. Decimal 7 ka binary kya hoga?
2. Hexadecimal ki base kitni hai?
3. Hex mein A ki qeemat kitni hai?
4. Bit kis ko kehte hain?
5. Bare A ka ASCII code kitna hai?

Drill Answers: 1. 111 2. 16 3. 10 4. Binary digit (0 ya 1) ko 5. 65

Chapter 10

Programming, Database aur Computer Security

Aakhri chapter teen aham ilaqon ko jorta hai - programming ki bunyaad (algorithm, flowchart), database ke usool (record, field, primary key) aur security (virus, firewall, phishing).

Notes - Asaan Alfaaz Mein

- **Algorithm kya hai:** kisi maslay ko hal karne ke **marhala-war (step-by-step) usool** ko algorithm kehte hain. Program likhne se pehle algorithm banaya jata hai - jese chai banane ki tarkeeb ek algorithm hai. Algorithm kisi bhi zuban mein likha ja sakta hai; use aasan farzi (fake) code mein likhna **Pseudocode** kehlata hai.
- **Flowchart:** algorithm ki **tasveeri shakal** ko flowchart kehte hain - is mein mukhtalif shaklein (shapes) mukhtalif kaam dikhati hain. Yaad rakhein: **Oval (anda numa)** = shuru aur aakhir (Start/End); **Parallelogram** = input aur output; **Rectangle** = processing (hisaab); **Diamond** = faisla (decision - haan/na wala sawal). Shapes ke ye jor test mein lazmi aate hain.
- **Programming Languages ki satah: Machine language** (0 aur 1) sab se neeche wali (low-level) hai - computer use seedha samajhta hai magar insaan ke liye mushkil. **Assembly language** chhote alfaaz (codes) istemal karti hai. **High-level languages** insani zuban se qareeb hoti hain - **C, C++, Java, Python** - inhein compiler/interpreter se machine language mein badalna padta hai. Har program ke likhne ke usool (grammar) ko **Syntax** kehte hain.
- **Database kya hai:** jura hua data jab **tables** ki shakal mein manzoom rakha jaye to woh database hai - jese school ke students ka record. Database ko banane, rakhne aur istemal karne wale software ko **DBMS (Database Management System)** kehte hain - misal: MS Access, MySQL, Oracle.
- **Field, Record, Table:** table ki **column** ko **Field** kehte hain (jese sirf "Naam" wala hissa), table ki **row** ko **Record** kehte hain (jese ek student ki puri maloomat), aur rows-columns ka pura majmua **Table** hai. Jo field har record ko baqiyon se **alag aur unique** pehchan de - jese Roll Number - use **Primary Key** kehte hain; primary key ki qeemat kabhi do records mein ek jesi nahi ho sakti aur khaali bhi nahi rehti.
- **Virus aur Malware: Computer Virus** ek nuqsan-deh program hai jo khud ko copy karke files aur dusre computers ko kharab karta hai. Tamam nuqsan-deh programs ko mila kar **Malware** kehte hain - is ki aqsaam: **Worm** (khud phailta hai), **Trojan** (achay program ke bhes mein chhupa), **Spyware** (chhup kar jasoosi karna). Hifazat ke liye **Antivirus** software istemal hota hai.
- **Firewall, Phishing, Hacking: Firewall** hamare network aur internet ke darmiyan ek security deewar hai jo mashkook data ko rok ti hai. **Phishing** dhoke wali email/website hai jo aap se password ya bank maloomat nikalti hai - ajnabi links par hamesha shak karein. Kisi ke computer mein baghair ijazat ghusna **Hacking** kehlata hai aur ye jurm hai.
- **Backup aur Password:** aham data ki **dusri copy** kisi aur jagah rakhna **Backup** kehlata hai - virus ya kharabi ki surat mein data wapas mil jata hai. Mazboot password lamba, mukhtalif (haroof, numbers, nishanaat mila kar) aur har account ke liye alag hona chahiye. Ye chhoti chhoti aadatain hi asli security hain.

Key Facts - Yaad Rakhne Wale Jor aur Farq

Box 1: Flowchart ki shaklein - jor yaad karein

Oval: Start / End (shuru aur aakhir).

Parallelogram: Input / Output (data lena aur dena).

Rectangle: Process (hisaab aur kaam).

Diamond: Decision (haan / na wala faisla).

Box 2: Database ki istilahat - ek misal

Table: Students ki puri fehrist.

Field (column): sirf Naam, sirf Roll No - ek qisam ki maloomat.

Record (row): ek student ki mukammal maloomat.

Primary Key: Roll Number - har student ka alag.

Aam Ghaltiyan - Common Mistakes

- Flowchart mein faisla (decision) DIAMOND se hota hai, rectangle se nahi - rectangle processing ke liye hai.
- Database mein ROW record hoti hai aur COLUMN field - dono ko ulta karna aam ghalti hai.
- Trojan khud ko nahi phailata, worm khud phailta hai - virus ki aqsaam mein ye farq poocha jata hai.
- Firewall antivirus NAHI hai - firewall network ki deewar hai, antivirus file-programs ki jaanch karta hai.

Practice MCQs

1. A step-by-step set of instructions written to solve a problem is called an:

(a) Syntax (b) Output (c) Compiler (d) Algorithm

2. The pictorial (diagram) representation of an algorithm is called a:

(a) Database (b) Spreadsheet (c) Flowchart (d) Web page

3. In a flowchart, the diamond shape is used for:

(a) Decision (b) Start (c) Input only (d) Printing

4. In a flowchart, the oval shape is used for:

(a) Decision (b) Processing (c) Storing data (d) Start and End

5. In a flowchart, the parallelogram shape is used for:

(a) Decision (b) Input and Output (c) Start and End (d) Connecting pages

6. Which of the following is a high-level programming language?

(a) Python (b) Machine language (c) Assembly language (d) Binary code

7. An organized collection of related data stored in tables is called a:

(a) Folder (b) Web page (c) Database (d) Slide

8. A row of a database table is called a:

(a) Record (b) Field (c) Table (d) Key

9. A column of a database table is called a:

(a) Record (b) File (c) Field (d) Report

10. The field that uniquely identifies each record in a table is called:

(a) Foreign data (b) Primary Key (c) Field name (d) Record key number

11. Software used to create and manage databases, such as MS Access, is called:

(a) Word processor (b) Web browser (c) Antivirus (d) DBMS

12. A harmful program that copies itself and damages files in a computer is called:

(a) Backup (b) Patch (c) Driver (d) Virus

13. Which program is used to protect a computer from viruses?

(a) Antivirus (b) MS Paint (c) Calculator (d) Notepad

14. A security system that acts as a wall between a private network and the internet is called:

(a) Modem (b) Router cable (c) Firewall (d) Web server

15. Fraud (fake) emails or websites that try to steal your password or bank details are called:

(a) Booting (b) Phishing (c) Browsing (d) Backup

16. Getting illegal, unauthorized access to someone's computer system is called:

(a) Phishing only (b) Hacking (c) Surfing (d) Chatting

17. A copy of important data kept at another place for safety is called:

(a) Virus (b) Cache (c) Backup (d) Cookie file

Answer Key - Chapter 10

1. (d) 2. (c) 3. (a) 4. (d) 5. (b) 6. (a) 7. (c) 8. (a) 9. (c) 10. (b) 11. (d) 12. (d) 13. (a) 14. (c) 15. (b) 16. (b) 17. (c)

Try Yourself - Quick Drills

1. Flowchart mein faislay (decision) ki shakl kon si hai?

2. Primary key kya hoti hai?

3. Computer virus kya hai?

4. Backup kyun rakha jata hai?

5. DBMS ki ek misal dein.

Drill Answers: 1. Diamond 2. Har record ki unique pehchan wali field, jese Roll Number 3. Ek nuqsan-deh program jo khud copy ho kar files kharab karta hai 4. Data kharab ya khatam hone ki surat mein wapas pane ke liye 5. MS Access (ya MySQL, Oracle)

Final Test

Mixed Practice Test

Ye test tamam chapters se mila jula hai - bilkul asal test ki tarah. Waqt note karein aur khud ko check karein.

1. Who is called the Father of the Computer?

(a) Bill Gates (b) Ada Lovelace (c) Tim Berners-Lee (d) Charles Babbage

2. The default file extension of a document in new MS Word is:

(a) .docx (b) .xlsx (c) .pptx (d) .docxx

3. 1 GB is equal to:

(a) 1000 MB (b) 100 MB (c) 1024 MB (d) 1024 KB

4. Which of the following is an output device?

(a) Keyboard (b) Scanner (c) Mouse (d) Monitor

5. Find and Replace in MS Word is opened by:

(a) Ctrl + H (b) Ctrl + F (c) Ctrl + R (d) Ctrl + P

6. The main job of DNS is to change:

(a) Files into folders (b) Domain names into IP addresses (c) Text into images (d) Emails into files

7. The binary equivalent of decimal 9 is:

(a) 1010 (b) 1100 (c) 1000 (d) 1001

8. The slide show in PowerPoint is started from the beginning by pressing:

(a) F7 (b) F1 (c) F5 (d) Shift + F1

9. A primary key in a database is used to:

(a) Delete all records (b) Uniquely identify each record (c) Print the table (d) Change field names

10. Which of the following is a search engine?

(a) Google (b) Google Chrome (c) Mozilla Firefox (d) Microsoft Edge

11. Which memory is non-volatile (keeps data without power)?

(a) RAM (b) ROM (c) Cache (d) Registers

12. FTP is a protocol used for:

(a) Transferring files (b) Sending printed letters (c) Playing videos (d) Making slides

13. First generation computers used which component?

(a) Transistors (b) IC chips (c) Microprocessors (d) Vacuum tubes

Answer Key - Mixed Practice Test

1. (d) 2. (a) 3. (c) 4. (d) 5. (a) 6. (b) 7. (d) 8. (c) 9. (b) 10. (a) 11. (b) 12. (a) 13. (d)

Quick Chart

Shortcut Keys Quick-Chart

Test se pehle ye chart ek nazar zaroor dekhein - woh shortcuts jo har paper mein aate hain, un ke kaam aur woh program jahan woh istemal hote hain, ek hi jagah diye gaye hain.

Shortcut	Kaam (Work)	Kahan (Where)
Ctrl + S	Save karna	Word, Excel, PowerPoint
Ctrl + P	Print karna	Word, Excel, PowerPoint
Ctrl + X / C / V	Cut / Copy / Paste	Tamam Office programs
Ctrl + Z	Undo - kiya hua kaam wapas	Word, Excel, PowerPoint
Ctrl + F / H	Find / Replace - lafz dhoondhna / badalna	MS Word
F7	Spelling aur grammar check	MS Word
F5	Slide show shuru se chalana	MS PowerPoint
Shift + F5	Slide show current slide se chalana	MS PowerPoint

Word List

Computer Glossary - Alfaz ke Matlab

Is list mein is book ki aham istilahat (terms) ke asaan matlab diye gaye hain. Jo lafz samajh na aaye, yahan se dekh lein.

- **Data:** kache (raw) facts aur figures - jese numbers aur naam.
- **Information:** process shuda data jo maani-daar ho aur kaam aa sake.
- **Hardware:** computer ke woh tamam hissay jinhein chhua ja sake - CPU, monitor, keyboard.
- **Software:** programs ka majmua jo computer ko batata hai ke kya karna hai.
- **CPU:** Central Processing Unit - computer ka dimagh, asal processing yahi hoti hai.
- **RAM:** Random Access Memory - aarzi memory, bijli band hone par khaali ho jati hai.
- **ROM:** Read Only Memory - mustaqil memory, BIOS is mein hota hai.
- **Operating System:** woh system software jo poore computer ko chalata aur manage karta hai - Windows, Linux.
- **Compiler:** poore program ko ek saath machine language mein badalne wala translator.
- **Internet:** duniya bhar ke computers ka sab se bara network.
- **Browser:** websites kholne wala program - Chrome, Firefox, Edge.
- **Search Engine:** internet par cheezein dhoondhne wali website - Google, Bing.
- **Network:** do ya zyada computers ka data share karne ke liye jura hua majmua.
- **LAN:** Local Area Network - chhoti jagah (building) ka network.
- **Database:** tables ki shakal mein manzoom, jura hua data ka majmua.
- **Virus:** nuqsan-deh program jo khud copy ho kar files aur system ko kharab karta hai.
- **Algorithm:** maslay ke hal ke marhala-war (step-by-step) usool.
- **Flowchart:** algorithm ki tasveeri (diagram wali) shakal.
- **Binary:** base 2 ka number system - sirf 0 aur 1 par mabni, computer ki andar ki zuban.
- **Byte:** 8 bits ka majmua - ek harf ya number rakhne ki unit.
- **Modem:** computer aur internet line ke darmiyan signal badalne wali device.
- **URL:** Uniform Resource Locator - har webpage ka makhsoos patta.
- **Backup:** aham data ki dusri copy jo mehfooz jagah rakhi jaye.

Exam Tips

- Full forms mein ek lafz badla hota hai (Universal ki jagah United) - har lafz ghor se parhein; ye sab se asan number hain.
- RAM volatile, ROM non-volatile - ye ek jor computer ke aadhe sawalon ki kunji hai.
- Units ki seedhi rattein: 8 bits = 1 Byte, 1024 Bytes = 1 KB, 1024 KB = 1 MB, 1024 MB = 1 GB. 1000 wala option hamesha jaal hai.
- Generations ki tartib pakki karein: Vacuum Tube, Transistor, IC, Microprocessor, AI - sawal isi tartib ko ulta karke aata hai.
- CD 700 MB, DVD 4.7 GB, Blu-ray 25 GB - teeno capacities aksar ek hi sawal ke options mein hoti hain.

- Compiler poora program ek saath, interpreter line by line - farq ek line mein yaad rakhein.
- Google search engine hai, Chrome browser - ye farq har dusre paper mein zaroor aata hai.
- Excel mein formula = se shuru; Word mein Find Ctrl+F, Replace Ctrl+H; PowerPoint mein show F5 - shortcuts ko program ke saath jor kar yaad karein.
- Database: row = record, column = field, unique pehchan = primary key. Flowchart: diamond = decision, oval = start/end.
- Binary conversion ki roz 5 minute practice karein - 1 se 16 tak ke numbers ke binary zehen mein pakke kar lein, sawal asan ho jayega.

Best of luck! Mehnat kabhi zaya nahi jati - roz ka thora sa parhna, kal ki bari kamyabi banta hai. - MR Academy